08/10/2026

Status update

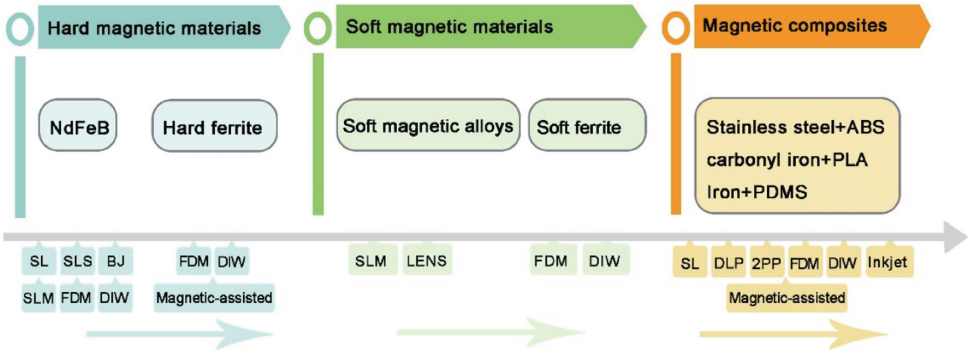
Agenda

Cover notable read research areas:

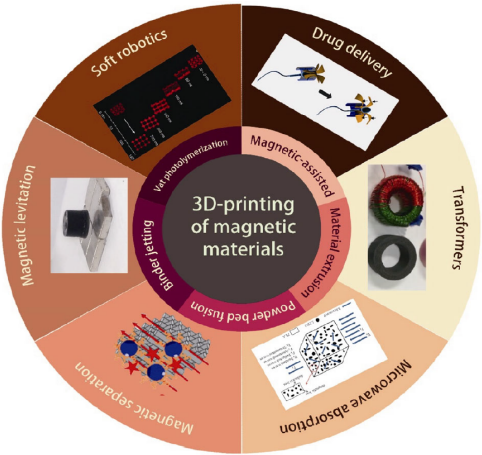
- **Review** - Xiangxia Wei et al. "Advances in 3D printing of magnetic materials: Fabrication, properties, and their applications"
- **Paper** - Lindsey M. Bollig et al. "3D printed magnetic polymer composite transformers"
- **Group** - Guo-Quan Lu's research

Propose a research direction
Clear up questions
Start formulating the project

Review paper - Chapter 4 - Methods



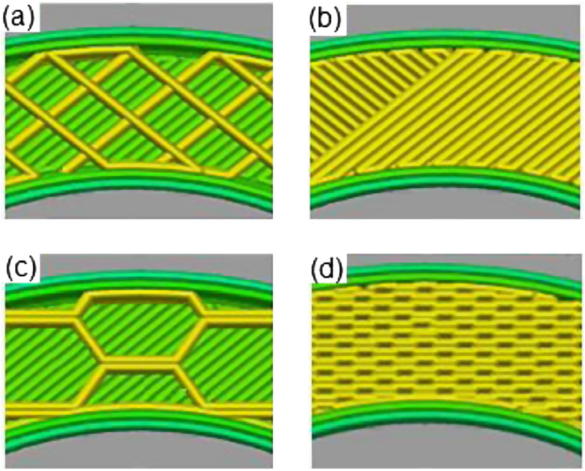
Review paper - Chapter 5 - Applications

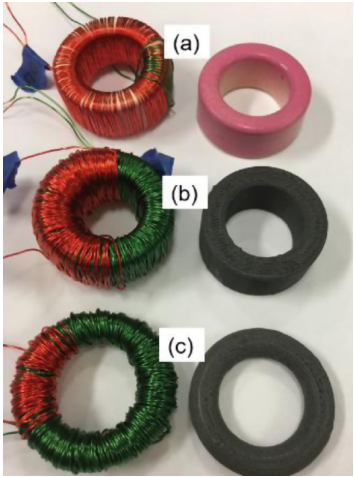


Review paper - Chapter 6 - Challanges

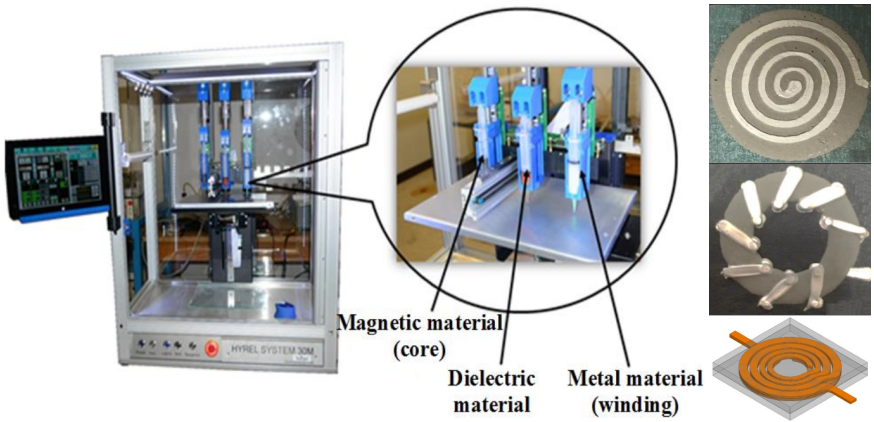
- It is necessary to explore geometries not feasible to be achieved by state-of-the-art methods, and have unique and potentially useful magnetic functionality.
- In PBF, powder properties need to be optimized and printing physics need to be studied.
- In FDM and DIW, the magnetic properties are difficult to precisely control due to a tradeoff in volume fraction of the magnetic filler.
- Homogeneous dispersion of magnetic fillers in the composites are a challenge.
- Further work could focus on in situ magnetic field alignment of particles for desired magnetic properties.

Paper - 3D printed magnetic composite transformers





Group - Guo-Quan Lu's research



Proposed research direction

Current research:

- Multi-permeability magnetic cores have been demonstrated
- Different materials lead to different permeability regions
- Shown to influence flux distribution and DC-bias behaviour

Proposed workflow:

- Simulate an inductor
- Vary μ in different regions
- Evaluate L, saturation, flux density, losses, DC-bias behaviour
- Optimize μ -distribution
- Can FDM reproduce it?

Questions

1. Is the end goal a fully printed inductor including the winding, or mainly printed magnetic components?
2. Why FDM/filament? Should we first define the engineering requirements and then choose the most suitable AM method?
3. Bima used superparamagnetic material, is this something we should consider?
4. How do you recommend I use AI? How can I use it to improve efficiency without relying on it for the core thinking and learning?